

OpenStax Calculus Volume 2

Section 5.5 Exercises: Alternating Series

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State whether each of the following series converges absolutely, conditionally, or not at all.

1. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n+3}$
2. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sqrt{n+1}}{\sqrt{n+3}}$
3. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n+3}}$
4. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sqrt{n+3}}{n}$
5. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n!}$
6. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3^n}{n!}$
7. $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{n-1}{n}\right)^n$
8. $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{n+1}{n}\right)^n$
9. $\sum_{n=1}^{\infty} (-1)^{n+1} \sin^2 n$
10. $\sum_{n=1}^{\infty} (-1)^{n+1} \cos^2 n$
11. $\sum_{n=1}^{\infty} (-1)^{n+1} \sin^2(1/n)$
12. $\sum_{n=1}^{\infty} (-1)^{n+1} \cos^2(1/n)$
13. $\sum_{n=1}^{\infty} (-1)^{n+1} \ln(1/n)$
14. $\sum_{n=1}^{\infty} (-1)^{n+1} \ln(1 + \frac{1}{n})$
15. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{1+n^4}$
16. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^e}{1+n^\pi}$
17. $\sum_{n=1}^{\infty} (-1)^{n+1} 2^{1/n}$
18. $\sum_{n=1}^{\infty} (-1)^{n+1} n^{1/n}$
19. $\sum_{n=1}^{\infty} (-1)^n (1 - n^{1/n})$ (Hint: $n^{1/n} \approx 1 + \ln(n)/n$ for large n .)
20. $\sum_{n=1}^{\infty} (-1)^{n+1} n(1 - \cos(\frac{1}{n}))$ (Hint: $\cos(1/n) \approx 1 - 1/n^2$ for large n .)

21. $\sum_{n=1}^{\infty} (-1)^{n+1} (\sqrt{n+1} - \sqrt{n})$ (Hint: Rationalize the numerator.)
22. $\sum_{n=1}^{\infty} (-1)^{n+1} (\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}})$ (Hint: Find common denominator then rationalize numerator.)
23. $\sum_{n=1}^{\infty} (-1)^{n+1} (\ln(n+1) - \ln n)$
24. $\sum_{n=1}^{\infty} (-1)^{n+1} n (\tan^{-1}(n+1) - \tan^{-1} n)$ (Hint: Use the Mean Value Theorem.)
25. $\sum_{n=1}^{\infty} (-1)^{n+1} ((n+1)^2 - n^2)$
26. $\sum_{n=1}^{\infty} (-1)^{n+1} (\frac{1}{n} - \frac{1}{n+1})$
27. $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n}$
28. $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^{1/n}}$
29. $\sum_{n=1}^{\infty} \frac{1}{n} \sin(\frac{n\pi}{2})$
30. $\sum_{n=1}^{\infty} \sin(n\pi/2) \sin(1/n)$

In each of the following problems, use the estimate $|R_N| \leq b_{N+1}$ to find a value of N that guarantees that the sum of the first N terms of the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ differs from the infinite sum by at most the given error. Calculate the partial sum S_N for this N .

31. [T] $b_n = 1/n$, error $< 10^{-5}$
32. [T] $b_n = 1/\ln(n)$, $n \geq 2$, error $< 10^{-1}$
33. [T] $b_n = 1/\sqrt{n}$, error $< 10^{-3}$
34. [T] $b_n = 1/2^n$, error $< 10^{-6}$
35. [T] $b_n = \ln(1 + \frac{1}{n})$, error $< 10^{-3}$
36. [T] $b_n = 1/n^2$, error $< 10^{-6}$

For the following exercises, indicate whether each of the following statements is true or false. If the statement is false, provide an example in which it is false.

37. If $b_n \geq 0$ is decreasing and $\lim_{n \rightarrow \infty} b_n = 0$, then $\sum_{n=1}^{\infty} (b_{2n-1} - b_{2n})$ converges absolutely.
38. If $b_n \geq 0$ is decreasing, then $\sum_{n=1}^{\infty} (b_{2n-1} - b_{2n})$ converges absolutely.
39. If $b_n \geq 0$ and $\lim_{n \rightarrow \infty} b_n = 0$, then $\sum_{n=1}^{\infty} (\frac{1}{2}(b_{3n-2} + b_{3n-1}) - b_{3n})$ converges.
40. If $b_n \geq 0$ is decreasing and $\sum_{n=1}^{\infty} (b_{3n-2} + b_{3n-1} - b_{3n})$ converges, then $\sum_{n=1}^{\infty} b_{3n-2}$ converges.
41. If $b_n \geq 0$ is decreasing and $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ converges conditionally but not absolutely, then b_n does not tend to zero.
42. Let $a_n^+ = a_n$ if $a_n \geq 0$ and $a_n^- = -a_n$ if $a_n < 0$. (Also, $a_n^+ = 0$ if $a_n < 0$ and $a_n^- = 0$ if $a_n \geq 0$.) If $\sum_{n=1}^{\infty} a_n$ converges conditionally but not absolutely, then neither $\sum_{n=1}^{\infty} a_n^+$ nor $\sum_{n=1}^{\infty} a_n^-$ converges.
43. Suppose that a_n is a sequence of positive real numbers and that $\sum_{n=1}^{\infty} a_n$ converges. Suppose that b_n is an arbitrary sequence of ones and minus ones. Does $\sum_{n=1}^{\infty} a_n b_n$ necessarily converge?
44. Suppose that a_n is a sequence such that $\sum_{n=1}^{\infty} a_n b_n$ converges for every possible sequence b_n of zeros and ones. Does $\sum_{n=1}^{\infty} a_n$ converge absolutely?

The following series do not satisfy the hypotheses of the alternating series test as stated.

In each case, state which hypothesis is not satisfied. State whether the series converges absolutely.

45. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin^2 n}{n}$
46. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\cos^2 n}{n}$
47. $1 + \frac{1}{2} - \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \frac{1}{6} - \frac{1}{7} - \frac{1}{8} + \cdots$
48. $1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} + \frac{1}{8} - \frac{1}{9} + \cdots$
49. Show that the alternating series $1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} + \frac{1}{4} - \frac{1}{8} + \cdots$ does not converge. What hypothesis of the alternating series test is not met?
50. Suppose that $\sum a_n$ converges absolutely. Show that the series consisting of the positive terms a_n also converges.
51. Show that the alternating series $\frac{2}{3} - \frac{3}{5} + \frac{4}{7} - \frac{5}{9} + \cdots$ does not converge. What hypothesis of the alternating series test is not met?
52. The formula $\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots$ will be derived in the next chapter. Use the remainder $|R_N| \leq b_{N+1}$ to find a bound for the error in estimating $\cos \theta$ by the fifth partial sum $1 - \theta^2/2! + \theta^4/4! - \theta^6/6! + \theta^8/8!$ for $\theta = 1$, $\theta = \pi/6$, and $\theta = \pi$.
53. The formula $\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots$ will be derived in the next chapter. Use the remainder $|R_N| \leq b_{N+1}$ to find a bound for the error in estimating $\sin \theta$ by the fifth partial sum $\theta - \theta^3/3! + \theta^5/5! - \theta^7/7! + \theta^9/9!$ for $\theta = 1$, $\theta = \pi/6$, and $\theta = \pi$.
54. How many terms in $\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots$ are needed to approximate $\cos 1$ accurate to an error of at most 0.00001?
55. How many terms in $\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots$ are needed to approximate $\sin 1$ accurate to an error of at most 0.00001?
56. Sometimes the alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ converges to a certain fraction of an absolutely convergent series $\sum_{n=1}^{\infty} b_n$ at a faster rate. Given that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, find $\frac{\pi^2}{12} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$. Which of the series $6 \sum_{n=1}^{\infty} \frac{1}{n^2}$ and $12 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$ gives a better estimation of π^2 using 1000 terms?

The following alternating series converge to given multiples of π . Find the value of N predicted by the remainder estimate such that the N th partial sum of the series accurately approximates the left-hand side to within the given error. Find the minimum N for which the error bound holds, and give the desired approximate value in each case. Up to 15 decimal places, $\pi = 3.141592653589793 \dots$

57. [T] $\frac{\pi}{4} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$, error < 0.0001
58. [T] $\frac{\pi}{\sqrt{12}} = \sum_{k=0}^{\infty} \frac{(-3)^{-k}}{2k+1}$, error < 0.0001
59. [T] The series $\sum_{n=0}^{\infty} \frac{\sin(x+\pi n)}{x+\pi n}$ plays an important role in signal processing. Show that $\sum_{n=0}^{\infty} \frac{\sin(x+\pi n)}{x+\pi n}$ converges whenever $0 < x < \pi$. (Hint: Use the formula for the sine of a sum of angles.)
60. [T] If $\sum_{n=1}^N (-1)^{n-1} \frac{1}{n} \rightarrow \ln 2$, what is $1 + \frac{1}{3} + \frac{1}{5} - \frac{1}{2} - \frac{1}{4} - \frac{1}{6} + \frac{1}{7} + \frac{1}{9} + \frac{1}{11} - \frac{1}{8} - \frac{1}{10} - \frac{1}{12} + \cdots$?

61. [T] Plot the partial sum $\sum_{n=1}^{100} \frac{\cos(2\pi nx)}{n}$ for $0 \leq x < 1$. Explain why $\sum_{n=1}^{\infty} \frac{\cos(2\pi nx)}{n}$ diverges when $x = 0, 1$. How does the series behave for other x ?
62. [T] Plot the series $\sum_{n=1}^{100} \frac{\sin(2\pi nx)}{n}$ for $0 \leq x < 1$ and comment on its behavior
63. [T] Plot the series $\sum_{n=1}^{100} \frac{\cos(2\pi nx)}{n^2}$ for $0 \leq x < 1$ and describe its graph.
64. [T] The alternating harmonic series converges because of cancellation among its terms. Its sum is known because the cancellation can be described explicitly. A random harmonic series is one of the form $\sum_{n=1}^{\infty} \frac{s_n}{n}$, where s_n is a randomly generated sequence of ± 1 's in which the values ± 1 are equally likely to occur. Use a random number generator to produce 1000 random ± 1 's and plot the partial sums $S_N = \sum_{n=1}^N \frac{s_n}{n}$ of your random harmonic sequence for $N = 1$ to 1000. Compare to a plot of the first 1000 partial sums of the harmonic series.
65. [T] Estimates of $\sum_{n=1}^{\infty} \frac{1}{n^2}$ can be accelerated by writing its partial sums as $\sum_{n=1}^N \frac{1}{n^2} = \sum_{n=1}^N \frac{1}{n(n+1)} + \sum_{n=1}^N \frac{1}{n^2(n+1)}$ and recalling that $\sum_{n=1}^N \frac{1}{n(n+1)} = 1 - \frac{1}{N+1}$ converges to one as $N \rightarrow \infty$. Compare the estimate of $\pi^2/6$ using the sums $\sum_{n=1}^{1000} \frac{1}{n^2}$ with the estimate using $1 + \sum_{n=1}^{1000} \frac{1}{n^2(n+1)}$.
66. [T] The Euler transform rewrites $S = \sum_{n=0}^{\infty} (-1)^n b_n$ as $S = \sum_{n=0}^{\infty} 2^{-n-1} \sum_{m=0}^n (-1)^m \binom{n}{m} b_m$. For the alternating harmonic series, it takes the form $\ln(2) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \sum_{n=1}^{\infty} \frac{1}{n2^n}$. Compute partial sums of $\sum_{n=1}^{\infty} \frac{1}{n2^n}$ until they approximate $\ln(2)$ accurate to within 0.0001. How many terms are needed? Compare this answer to the number of terms of the alternating harmonic series needed to estimate $\ln(2)$.
67. [T] In the text it was stated that a conditionally convergent series can be rearranged to converge to any number. Here is a slightly simpler, but similar, fact. If $a_n \geq 0$ is such that $a_n \rightarrow 0$ as $n \rightarrow \infty$ but $\sum_{n=1}^{\infty} a_n$ diverges, then, given any number A there is a sequence s_n of ± 1 's such that $\sum_{n=1}^{\infty} a_n s_n \rightarrow A$. Show this for $A > 0$ as follows.
- Recursively define s_n by $s_n = 1$ if $S_{n-1} = \sum_{k=1}^{n-1} a_k s_k < A$ and $s_n = -1$ otherwise.
 - Explain why eventually $S_n \geq A$, and for any m larger than this n , $A - a_m \leq S_m \leq A + a_m$.
 - Explain why this implies that $S_n \rightarrow A$ as $n \rightarrow \infty$.